

Euler's Principal Rotation

(Cf. Hanspeter Schaub, and John L. Junkins, "Analytical Mechanics of Space Systems," p.87~95)

Principal Rotation Theorem;

A rigid body can be brought from an arbitrary initial orientation to an arbitrary final orientation by a single rotation of the body through a principal angle(ϕ) about a principal line ($\hat{l}$); the principal line being a judicious axis ($\hat{l}$) fixed in the body and fixed in space.

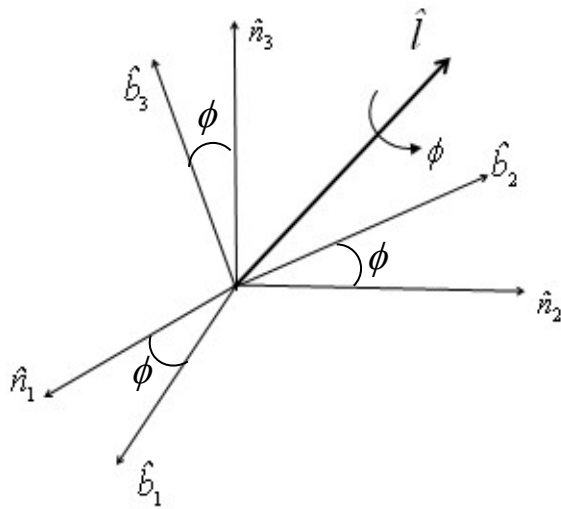
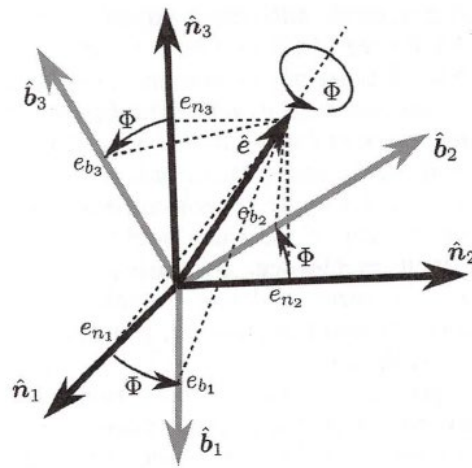


Figure 1.



a supplement figure of Figure 1.

$\vec{r}$ is displacement vector from body-frame point O to an arbitrary body-fixed point P. Upon rotating the body through an angle ϕ it is clear that P will be displaced along a circle (of radius $(r \sin \theta)$) to a position P'.

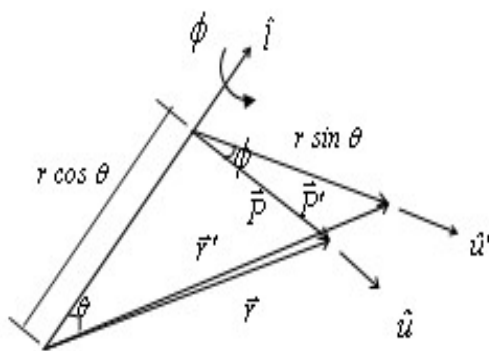
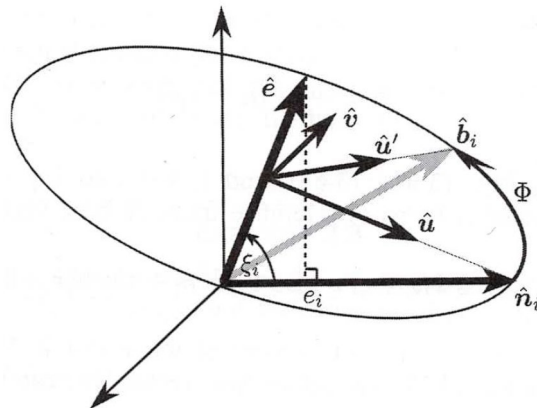


Figure 2.



a supplement figure of Figure 2.

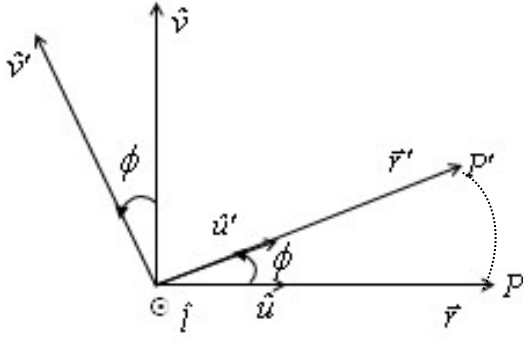


Figure 3

From Figure 2 and Figure 3, we obtain

$$\vec{r} = r \cos \theta \hat{l} + r \sin \theta \hat{u} \quad (1)$$

$$\vec{r}' = r \cos \theta \hat{l} + r \sin \theta \hat{u}' \quad (2)$$

$$\hat{u}' = \cos \phi \hat{u} + \sin \phi \hat{v} \quad (3)$$

$$\hat{v} = \frac{\hat{l} \times \vec{r}}{|\hat{l} \times \vec{r}|} = \frac{1}{r \sin \theta} \hat{l} \times \vec{r} \quad (4)$$

$$\hat{u} = \hat{v} \times \hat{l} = \frac{1}{r \sin \theta} (\hat{l} \times \vec{r}) \times \hat{l} \quad \text{use } \vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$$

$$\text{use } \vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c}$$

Equation (5) reduces to

$$\hat{u} = \hat{v} \times \hat{l} = -\frac{1}{r \sin \theta} [\hat{l} \times (\hat{l} \times \vec{r})]$$

$$\text{where } \hat{l} \times (\hat{l} \times \vec{r}) = (\hat{l} \cdot \vec{r}) \hat{l} - (\hat{l} \cdot \hat{l}) \vec{r} = (\hat{l} \cdot \vec{r}) \hat{l} - \vec{r}$$

$$\hat{u} = \frac{1}{r \sin \theta} [\vec{r} - (\hat{l} \cdot \vec{r}) \hat{l}] \quad (6)$$

Substitution of Equation (6), (4), (3) into Eq.(2) yields an important geometrical results.

$$\begin{aligned} \vec{r}' &= r \cos \theta \hat{l} + r \sin \theta \hat{u}' = r \cos \theta \hat{l} + r \sin \theta (\cos \phi \hat{u} + \sin \phi \hat{v}) \\ &= (\vec{r} \cdot \hat{l}) \hat{l} + r \sin \theta \left[\cos \phi \left(\frac{1}{r \sin \theta} (\vec{r} - (\hat{l} \cdot \vec{r}) \hat{l}) \right) + \sin \phi \frac{1}{r \sin \theta} (\hat{l} \times \vec{r}) \right] \\ \vec{r}' &= (1 - \cos \phi) (\hat{l} \cdot \vec{r}) \hat{l} + \cos \phi \vec{r} + \sin \phi (\hat{l} \times \vec{r}) \end{aligned} \quad (7)$$

Eq.(7) is the general vector equation for the space curve generated by an arbitrary point P fixed in a rigid body, rotating about a fixed axis.

Letting (Figure 1) the body fixed axes $\{\hat{b}\}$ be “initially coincident with fixed axis $\{\hat{n}\}$ ”, we can use Euler’s principal rotation theorem to develop several elegant parameterizations of the direction cosine matrix $[C]$ defining $\{\hat{b}\}$ ’s instantaneous angular position in the sense.

$$\{\hat{b}\}=[C]\{\hat{n}\} \quad (8)$$

Let us denote

$$\hat{l} = l_{b1}\hat{b}_1 + l_{b2}\hat{b}_2 + l_{b3}\hat{b}_3 \quad (9)$$

$$\hat{l} = l_{n1}\hat{n}_1 + l_{n2}\hat{n}_2 + l_{n3}\hat{n}_3 \quad (10)$$

As a direct consequence of the fact that l_{bi} and l_{ni} are constant (during a rotation ϕ about a fixed $\hat{l}$) and $\hat{b}_i$ are initially coincident with $\hat{n}_i$.

$$l_{ni} = l_{bi} = l_i \quad (i=1,2,3)$$

If we take $\vec{r} = \hat{n}_i$ and $\vec{r}' = \hat{n}_i' \equiv \hat{b}_i$ in the general eq. (7), we obtain

$$\hat{b}_i = (1 - \cos \phi)(\hat{l} \cdot \hat{n}_i)\hat{l} + \cos \phi \hat{n}_i + \sin \phi(\hat{l} \times \hat{n}_i) \quad (11)$$

$$\begin{aligned} \Rightarrow \hat{b}_1 &= (1 - \cos \phi)l_1(l_1\hat{n}_1 + l_2\hat{n}_2 + l_3\hat{n}_3) + \cos \phi \hat{n}_1 + \sin \phi(l_3\hat{n}_2 - l_2\hat{n}_3) \\ &= [l_1^2(1 - \cos \phi) + \cos \phi]\hat{n}_1 + [l_1l_2(1 - \cos \phi) + \sin \phi l_3]\hat{n}_2 + [l_1l_3(1 - \cos \phi) - \sin \phi l_2]\hat{n}_3 \\ \Rightarrow \hat{b}_2 &= (1 - \cos \phi)l_2(l_1\hat{n}_1 + l_2\hat{n}_2 + l_3\hat{n}_3) + \cos \phi \hat{n}_2 + \sin \phi(l_1\hat{n}_3 - l_3\hat{n}_1) \\ &= [l_1l_2(1 - \cos \phi) + l_3 \sin \phi]\hat{n}_1 + [l_2^2(1 - \cos \phi) + \cos \phi]\hat{n}_2 + [l_2l_3(1 - \cos \phi) + \sin \phi l_1]\hat{n}_3 \\ \Rightarrow \hat{b}_3 &= (1 - \cos \phi)l_3(l_1\hat{n}_1 + l_2\hat{n}_2 + l_3\hat{n}_3) + \cos \phi \hat{n}_3 + \sin \phi(l_2\hat{n}_1 - l_1\hat{n}_2) \\ &= [l_1l_3(1 - \cos \phi) + l_2 \sin \phi]\hat{n}_1 + [l_2l_3(1 - \cos \phi) - l_1 \sin \phi]\hat{n}_2 + [l_3^2(1 - \cos \phi) + \cos \phi]\hat{n}_3 \end{aligned}$$

Then, we have

$$\begin{bmatrix} \hat{b}_1 \\ \hat{b}_2 \\ \hat{b}_3 \end{bmatrix} = [C] \begin{bmatrix} \hat{n}_1 \\ \hat{n}_2 \\ \hat{n}_3 \end{bmatrix}$$

$$[C] = \begin{bmatrix} l_1^2(1 - \cos \phi) + \cos \phi & l_1 l_2(1 - \cos \phi) + l_3 \sin \phi & l_1 l_3(1 - \cos \phi) - l_2 \sin \phi \\ l_2 l_1(1 - \cos \phi) - l_3 \sin \phi & l_2^2(1 - \cos \phi) + \cos \phi & l_2 l_3(1 - \cos \phi) + l_1 \sin \phi \\ l_3 l_1(1 - \cos \phi) + l_2 \sin \phi & l_3 l_2(1 - \cos \phi) - l_1 \sin \phi & l_3^2(1 - \cos \phi) + \cos \phi \end{bmatrix} \quad (12)$$

$$(l_1^2 + l_2^2 + l_3^2 = 1)$$

From equation (12),

$$\cos \phi = \frac{1}{2}(C_{11} + C_{22} + C_{33} - 1) \quad (13)$$

By differencing the symmetric elements, we have

$$\begin{aligned} 2l_3 \sin \phi &= C_{12} - C_{21} \\ 2l_2 \sin \phi &= C_{31} - C_{13} \\ 2l_1 \sin \phi &= C_{23} - C_{32} \end{aligned} \quad (14)$$

multiplying the above equations by l_3, l_2, l_1 respectively, we have

$$\sin \phi = \frac{1}{2}[l_3(C_{12} - C_{21}) + l_2(C_{31} - C_{13}) + l_1(C_{23} - C_{32})] \quad (15)$$

If [C] is given,

We can solve for $\hat{l}$ from equation(8)

$$\{\hat{b}\} = [C] \{\hat{n}\} \quad \text{and} \quad l_{ni} = l_{bi} = l_i \quad (i = 1, 2, 3)$$

$$\begin{bmatrix} l_1 \\ l_2 \\ l_3 \end{bmatrix} = [C] \begin{bmatrix} l_1 \\ l_2 \\ l_3 \end{bmatrix} \quad (16)$$

$$[C]\hat{l} = \hat{l}$$

$\hat{l}$ is the corresponding unit eigenvector of [C] that (eigenvalue)=1.

From eq (13) & eq (15) we can get ϕ with (l_3, l_2, l_1) from eq. (16)

Euler parameters (Quaternions)

We have four Euler parameters in terms of the principal line($\hat{l}$) principal angle (ϕ). In other words, the Euler parameter vector is defined in terms of the principal rotation elements as:

$$\begin{aligned} q_1 &= l_1 \sin \frac{\phi}{2} \\ q_2 &= l_2 \sin \frac{\phi}{2} \\ q_3 &= l_3 \sin \frac{\phi}{2} \\ q_4 &= \cos \frac{\phi}{2} \end{aligned} \tag{1}$$

It is obvious that the four q_i 's satisfy the constraint equation :

$$q_1^2 + q_2^2 + q_3^2 + q_4^2 = 1 \tag{2}$$

Let us use

$$\begin{aligned} \sin \phi &= 2 \sin \frac{\phi}{2} \cos \frac{\phi}{2} \\ \cos \phi &= 2 \cos^2 \frac{\phi}{2} - 1 \\ 1 - \cos \phi &= 2 \sin^2 \frac{\phi}{2} \end{aligned} \tag{3}$$

$$\begin{aligned} C_{11} &= l_1^2 (1 - \cos \phi) + \cos \phi = 2l_1^2 \sin^2 \frac{\phi}{2} + 2 \cos^2 \frac{\phi}{2} - 1 = 2q_1^2 + 2q_4^2 - q_1^2 - q_2^2 - q_3^2 - q_4^2 \\ &= q_1^2 - q_2^2 - q_3^2 + q_4^2 \end{aligned}$$

$$C_{12} = l_1 l_2 (1 - \cos \phi) + l_3 \sin \phi = 2l_1 l_2 \sin^2 \frac{\phi}{2} + l_3 2 \sin \frac{\phi}{2} \cos \frac{\phi}{2} = 2q_1 q_2 + 2q_4 q_3 = 2(q_1 q_2 + q_4 q_3)$$

$$C_{13} = l_1 l_3 (1 - \cos \phi) - l_2 \sin \phi = 2l_1 l_3 \sin^2 \frac{\phi}{2} - l_2 2 \sin \frac{\phi}{2} \cos \frac{\phi}{2} = 2q_1 q_3 - 2q_4 q_2 = 2(q_1 q_3 - q_4 q_2)$$

$$C_{21} = l_1 l_2 (1 - \cos \phi) - l_3 \sin \phi = 2l_1 l_2 \sin^2 \frac{\phi}{2} - l_3 2 \sin \frac{\phi}{2} \cos \frac{\phi}{2} = 2q_1 q_2 - 2q_4 q_3 = 2(q_1 q_2 - q_4 q_3)$$

$$\begin{aligned} C_{22} &= l_2^2 (1 - \cos \phi) + \cos \phi = 2l_2^2 \sin^2 \frac{\phi}{2} + 2 \cos^2 \frac{\phi}{2} - 1 = 2q_2^2 + 2q_4^2 - q_1^2 - q_2^2 - q_3^2 - q_4^2 \\ &= -q_1^2 + q_2^2 - q_3^2 + q_4^2 \end{aligned}$$

$$C_{23} = l_2 l_3 (1 - \cos \phi) + l_1 \sin \phi = 2l_2 l_3 \sin^2 \frac{\phi}{2} + l_1 2 \sin \frac{\phi}{2} \cos \frac{\phi}{2} = 2q_2 q_3 + 2q_4 q_1 = 2(q_2 q_3 + q_4 q_1)$$

$$\begin{aligned}
C_{31} &= l_1 l_3 (1 - \cos \phi) + l_2 \sin \phi = 2l_1 l_3 \sin^2 \frac{\phi}{2} + l_2 2 \sin \frac{\phi}{2} \cos \frac{\phi}{2} = 2q_1 q_3 + 2q_4 q_2 = 2(q_1 q_3 + q_4 q_2) \\
C_{32} &= l_2 l_3 (1 - \cos \phi) - l_1 \sin \phi = 2l_2 l_3 \sin^2 \frac{\phi}{2} - l_1 2 \sin \frac{\phi}{2} \cos \frac{\phi}{2} = 2q_2 q_3 - 2q_4 q_1 = 2(q_2 q_3 - q_4 q_1) \\
C_{33} &= l_3^2 (1 - \cos \phi) + \cos \phi = 2l_3^2 \sin^2 \frac{\phi}{2} + 2 \cos^2 \frac{\phi}{2} - 1 = 2q_3^2 + 2q_4^2 - q_1^2 - q_2^2 - q_3^2 - q_4^2 \\
&= -q_1^2 - q_2^2 + q_3^2 + q_4^2
\end{aligned}$$

The direction cosine matrix can be parameterized as a function of the Euler parameters:

$$[C(q_1, q_2, q_3, q_4)]$$

$$C = \begin{bmatrix} q_1^2 - q_2^2 - q_3^2 + q_4^2 & 2(q_1 q_2 + q_3 q_4) & 2(q_1 q_3 - q_2 q_4) \\ 2(q_1 q_2 - q_3 q_4) & -q_1^2 + q_2^2 - q_3^2 + q_4^2 & 2(q_2 q_3 + q_1 q_4) \\ 2(q_1 q_3 + q_2 q_4) & 2(q_2 q_3 - q_1 q_4) & -q_1^2 - q_2^2 + q_3^2 + q_4^2 \end{bmatrix} \quad (4)$$

The inverse relationship can be reduced from eq. (4).

$$\begin{aligned}
q_1 &= (C_{23} - C_{32}) / 4q_4 \\
q_2 &= (C_{31} - C_{13}) / 4q_4 \\
q_3 &= (C_{12} - C_{21}) / 4q_4 \\
q_4 &= \pm \frac{1}{2} (1 + C_{11} + C_{22} + C_{33})^{1/2}
\end{aligned} \quad (5)$$

+ sign can be used without loss of generality $\leftrightarrow$ changing all four signs of the q_i in eq. (4) does not change the direction cosine matrix.

Other terms are

$$\begin{aligned}
q_2 q_3 &= (C_{23} + C_{32}) / 4 \\
q_3 q_1 &= (C_{31} + C_{13}) / 4 \\
q_1 q_2 &= (C_{12} + C_{21}) / 4
\end{aligned} \quad (5')$$

If we know $q \rightarrow [C] \rightarrow$ 3-1-3 Euler angles

$$\begin{aligned}
\theta &= \cos^{-1}(C_{33}) \\
\phi &= \tan^{-1}(C_{31} / -C_{32}) \\
\psi &= \cos^{-1}(C_{13} / C_{23})
\end{aligned} \quad (6)$$

Euler parameter가 rotation matrix를 통해 Euler angle과 관계가 있다.
i.e., Euler parameter $\leftrightarrow$ Euler angle

We seek differential equation of form : $\dot{q} = \text{function}(q_1, q_2, q_3, q_4, \omega_1, \omega_2, \omega_3)$

These can be obtained by differentiating eq. (5).

For example,

$$q_4^2 = \frac{1}{4}(1 + C_{11} + C_{22} + C_{33})$$

$$\rightarrow 2q_4\dot{q}_4 = \frac{1}{4}(\dot{C}_{11} + \dot{C}_{22} + \dot{C}_{33}) \Rightarrow \dot{q}_4 = \frac{(\dot{C}_{11} + \dot{C}_{22} + \dot{C}_{33})}{8q_4}$$

$$\text{and from } [\dot{C}] = -[\tilde{\omega}][C] = -\begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix} \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}$$

$$\dot{C}_{11} = \omega_3 C_{21} - \omega_2 C_{31}$$

$$\dot{C}_{22} = \omega_1 C_{32} - \omega_3 C_{12}$$

$$\dot{C}_{33} = \omega_2 C_{13} - \omega_1 C_{23}$$

$$\begin{aligned} \Rightarrow \dot{q}_4 &= \frac{1}{8q_4} [(C_{32} - C_{23})\omega_1 + (C_{13} - C_{31})\omega_2 + (C_{21} - C_{12})\omega_3] \quad \text{using eq(5) and eq(5')} \\ &= \frac{1}{8q_4} [-4q_1q_4\omega_1 - 4q_2q_4\omega_2 - 4q_3q_4\omega_3] = -\frac{1}{2}(q_1\omega_1 + q_2\omega_2 + q_3\omega_3) \end{aligned}$$

Similarly, we obtain $\dot{q}_1, \dot{q}_2, \dot{q}_3$. Then, we have

$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \dot{q}_4 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 0 & \omega_3 & -\omega_2 & \omega_1 \\ -\omega_3 & 0 & \omega_1 & \omega_2 \\ \omega_2 & -\omega_1 & 0 & \omega_3 \\ -\omega_1 & -\omega_2 & -\omega_3 & 0 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix} \quad (7)$$

$$\text{or } \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \dot{q}_4 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} q_4 & -q_3 & q_2 & q_1 \\ q_3 & q_4 & -q_1 & q_2 \\ -q_2 & q_1 & q_4 & q_3 \\ -q_1 & -q_2 & -q_3 & q_4 \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \\ 0 \end{bmatrix} \quad (8)$$

No singularities $\rightarrow$ attractive in computer simulation.

Define the rotation unambiguously up to two revolutions of the body.

(cf. Direction cosine cannot determine whether the body has actually performed a complete revolution or is still in its original position)

Angular momentum for one particle

The angular momentum ($\vec{H}$) is defined as

$$\vec{H} = \vec{r} \times (m\dot{\vec{r}})$$

$$\vec{F} = m\ddot{\vec{r}}$$

$$\rightarrow \vec{r} \times \vec{F} = \vec{r} \times (m\ddot{\vec{r}})$$

$$\frac{^i d}{dt} (\vec{r} \times \dot{\vec{r}}) = \dot{\vec{r}} \times \dot{\vec{r}} + \vec{r} \times \ddot{\vec{r}}$$

$$\begin{aligned} \Rightarrow \vec{r} \times \vec{F} &= \frac{^i d}{dt} (\vec{r} \times m\dot{\vec{r}}) \\ &= \frac{^i d}{dt} (\vec{H}) \end{aligned}$$

$$\Rightarrow \vec{r} \times \vec{F} = \vec{M} \quad (\text{torque})$$

$$\vec{M} = \frac{^i d}{dt} \vec{H} \quad : \text{The key to writing the eqs. of motion for rigid bodies.}$$

$\vec{r}$ 의 origin: Center of Mass of the body
 i.e., 힘이 물체에 질량 중심에 전달되지 않으면
 torque 발생 $\rightarrow$ 회전

If external torques are absent, the angular momentum is conserved.

$$\text{If } \vec{M} = 0 \rightarrow \vec{H} = \text{const.}$$

Angular momentum of Rigid body

$$\vec{H} = \sum_{i=1}^n (\vec{r}_i \times m_i \dot{\vec{r}}_i) \quad ; \text{ for system of particles}$$

$$\vec{H} = \int (\vec{r} \times \dot{\vec{r}}) dm \quad ; \text{ for a rigid body}$$

The total angular momentum is the sum of the angular momentum for each mass element (dm). The integral extends over the entire object. $\dot{\vec{r}}$ is the inertial velocity of the mass element (dm) about the center of mass.

$\vec{r}$ of mass element changes with time in the inertial frame. $\rightarrow$ inconvenient

If do these integrals in a frame of reference tied to the body, it is convenient, because the mass distribution is static in such a frame.

$$\vec{v} = \vec{v}^B + \vec{\omega}^{bi} \times \vec{r} \quad , \quad \vec{\omega}^{bi} \text{ is the inertial angular velocity of the rigid body.}$$

$$\vec{H} = \int \vec{r} \times (\vec{\omega}^{bi} \times \vec{r}) dm \quad \vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c}$$

$$\vec{r} = x\hat{b}_1 + y\hat{b}_2 + z\hat{b}_3 \quad , \quad \vec{\omega}^{bi} = \omega_x \hat{b}_1 + \omega_y \hat{b}_2 + \omega_z \hat{b}_3$$

$$\vec{H} = H_x \hat{b}_1 + H_y \hat{b}_2 + H_z \hat{b}_3 \quad \text{expressed in body-frame.}$$

$$\begin{aligned} \rightarrow \dot{H} = & \left[\int (y^2 + z^2) dm \omega_x - \int xy dm \omega_y - \int xz dm \omega_z \right] \hat{b}_1 \\ & + \left[- \int xy dm \omega_x + \int (x^2 + z^2) dm \omega_y - \int yz dm \omega_z \right] \hat{b}_2 \\ & + \left[- \int xz dm \omega_x - \int yz dm \omega_y + \int (x^2 + y^2) dm \omega_z \right] \hat{b}_3 \end{aligned}$$

Let

$$\begin{aligned} I_{xy} &= - \int xy dm = I_{yx} \quad , & I_{xz} &= - \int xz dm = I_{zx} \quad , & I_{yz} &= - \int yz dm = I_{zy} \\ I_{xx} &= \int (y^2 + z^2) dm \quad , & I_{yy} &= \int (x^2 + z^2) dm \quad , & I_{zz} &= \int (x^2 + y^2) dm \end{aligned}$$

$$\begin{bmatrix} H_x \\ H_y \\ H_z \end{bmatrix} = \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{bmatrix} \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} \quad \leftrightarrow \quad \vec{H} = [I] \vec{\omega}^{bi} \quad ,$$

where $[I]$ is a symmetric matrix.

The angular momentum and the angular velocity are not generally aligned.

Principal axis of inertia

Q; When is $\vec{H}$ parallel to $\vec{\omega}$?

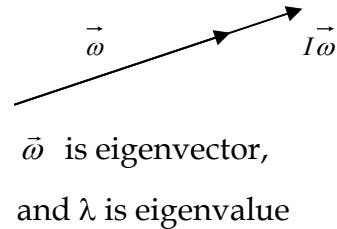
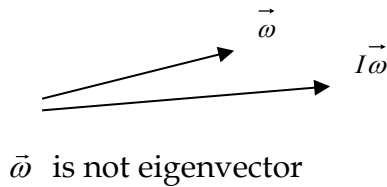
A; $\vec{H} = I\vec{\omega} = \lambda\vec{\omega}$, λ is scalar.

To find possible value of the scalar we have to solve

$$\{I - \lambda[1]\}\vec{\omega} = 0, \quad \vec{\omega} \neq 0$$

This is the eigenvalue problem. The λ value can be obtained as eigenvalues of I.

The eigenvectors will give the direction of $\vec{\omega}$ such that $\vec{H}$ and $\vec{\omega}$ are parallel.



λ is solution to $\det\{I - \lambda[1]\} = 0$

$\det\{I - \lambda[1]\} = 0$; characteristic equation $[I]; 3 \times 3 \text{ matrix} \rightarrow 3 \lambda_i \rightarrow 3 \omega_i$

I is symmetric $\rightarrow$ the eigenvector $\vec{\omega}_i$ corresponding to the three λ_i are perpendicular.

Proof.)

$$I\vec{\omega}_A = \lambda_A \vec{\omega}_A$$

$$I\vec{\omega}_B = \lambda_B \vec{\omega}_B$$

$$\rightarrow \vec{\omega}_B^T I\vec{\omega}_A = \vec{\omega}_B^T \lambda_A \vec{\omega}_A$$

$$- \vec{\omega}_A^T I\vec{\omega}_B = \vec{\omega}_A^T \lambda_B \vec{\omega}_B$$

$$\Rightarrow \vec{\omega}_B^T I\vec{\omega}_A - \vec{\omega}_A^T I\vec{\omega}_B = \vec{\omega}_B^T \lambda_A \vec{\omega}_A - \vec{\omega}_A^T \lambda_B \vec{\omega}_B$$

$$0 = \lambda_A \vec{\omega}_B^T \vec{\omega}_A - \lambda_B \vec{\omega}_A^T \vec{\omega}_B$$

$$0 = (\lambda_A - \lambda_B) \vec{\omega}_B \cdot \vec{\omega}_A$$

$$\vec{\omega}_B \cdot \vec{\omega}_A = 0 \quad (\because \lambda_1 \neq \lambda_2)$$

$\therefore \vec{\omega}_1$ and $\vec{\omega}_2$ are orthogonal.

There are only three possible axes in rigid body about which $\vec{H}$ and $\vec{\omega}$ are parallel.

The three special λ_i are called the principal moment of inertia, $\vec{\omega}_i$ called the principal axes of inertia.

Since three unit vector $(\hat{b}_1 = \frac{\vec{\omega}_A}{|\vec{\omega}_A|}, \hat{b}_2 = \frac{\vec{\omega}_B}{|\vec{\omega}_B|}, \hat{b}_3 = \frac{\vec{\omega}_C}{|\vec{\omega}_C|})$ are mutually perpendicular, they can be used as the basis vectors for a body frame of references.

This special body frame is called to principal axis frame.

Because $I\vec{\omega} = \lambda\vec{\omega}$ must be true when each of the unit vectors of this principal-axis frame is inserted, the moment of inertia tensor becomes

$$I = \begin{bmatrix} \lambda_A & 0 & 0 \\ 0 & \lambda_B & 0 \\ 0 & 0 & \lambda_C \end{bmatrix}$$

Principal axis
→[I] is become diagonalized

Note that I is diagonal $\vec{H}$ and $\vec{\omega}$ will be aligned only when the body rotates about one of the three principal axes. The principal axis frame has significant advantages for doing dynamics of rigid bodies. Since this frame can be always be found, we will usually work in it from now on.

Rotational kinetic energy of rigid body

$$\begin{aligned}
 T &= \int \frac{1}{2} \vec{v} \cdot \vec{v} dm \\
 &= \frac{1}{2} \int \left(\vec{\omega}^{bi} \times \vec{r} \right) \cdot \left(\vec{\omega}^{bi} \times \vec{r} \right) dm \\
 &= \frac{1}{2} \int \vec{\omega}^{bi} \cdot \left\{ \vec{r} \times \left(\vec{\omega}^{bi} \times \vec{r} \right) \right\} dm \\
 &= \frac{1}{2} \vec{\omega}^{bi} \cdot \underbrace{\int \vec{r} \times \left(\vec{\omega}^{bi} \times \vec{r} \right) dm}_{= \vec{H}}
 \end{aligned}$$

$$= \frac{1}{2} \vec{\omega}^{bi} \cdot \vec{H}$$

$$= \frac{1}{2} \vec{\omega}^{bi} \cdot [I] \vec{\omega}^{bi}$$

$\vec{v}$; inertial velocity $= {}^b \vec{v} + \left(\vec{\omega}^{bi} \times \vec{r} \right)$

$(\vec{A} \times \vec{B}) \cdot \vec{C} = \vec{A} \cdot (\vec{B} \times \vec{C})$

Euler's Equation

The equation of motion for a rigid body.

$$\vec{M} = \frac{d}{dt} \vec{H}$$

The applied torque must equal the inertial rate of change of the angular momentum.

$$\begin{aligned} \vec{H} &= I \vec{\omega}^{bi} && \text{body frame component} \\ \vec{M} &= \frac{d}{dt} I \vec{\omega}^{bi} + \vec{\omega}^{bi} \times I \vec{\omega}^{bi} && \leftarrow \dot{I} \vec{\omega}^{bi} = 0, (\because I = \text{constant in body frame}) \\ &= I \dot{\vec{\omega}} + \vec{\omega}^{bi} \times I \vec{\omega}^{bi} && \text{Expressed in body frame} \end{aligned}$$

$$I = \begin{bmatrix} A & 0 & 0 \\ 0 & B & 0 \\ 0 & 0 & C \end{bmatrix}$$

The moment of inertia matrix is simple when expressed in principal axis frame

Let $\vec{\omega}^{bi} = \omega_1 \hat{b}_1 + \omega_2 \hat{b}_2 + \omega_3 \hat{b}_3$ in principal axis body frame.

$$\begin{bmatrix} M_1 \\ M_2 \\ M_3 \end{bmatrix} = \begin{bmatrix} A & 0 & 0 \\ 0 & B & 0 \\ 0 & 0 & C \end{bmatrix} \begin{bmatrix} \dot{\omega}_1 \\ \dot{\omega}_2 \\ \dot{\omega}_3 \end{bmatrix} + \begin{bmatrix} \hat{b}_1 & \hat{b}_2 & \hat{b}_3 \\ \omega_1 & \omega_2 & \omega_3 \\ A\omega_1 & B\omega_2 & C\omega_3 \end{bmatrix}$$

$$\begin{aligned} \therefore M_1 &= A\dot{\omega}_1 + (C - B)\omega_2\omega_3 \\ M_2 &= B\dot{\omega}_2 + (A - C)\omega_1\omega_3 \\ M_3 &= C\dot{\omega}_3 + (B - A)\omega_1\omega_2 \end{aligned} \quad ; \text{Euler's equation}$$

They are three coupled nonlinear first-order differential equation one-half of rotational equations of motions.